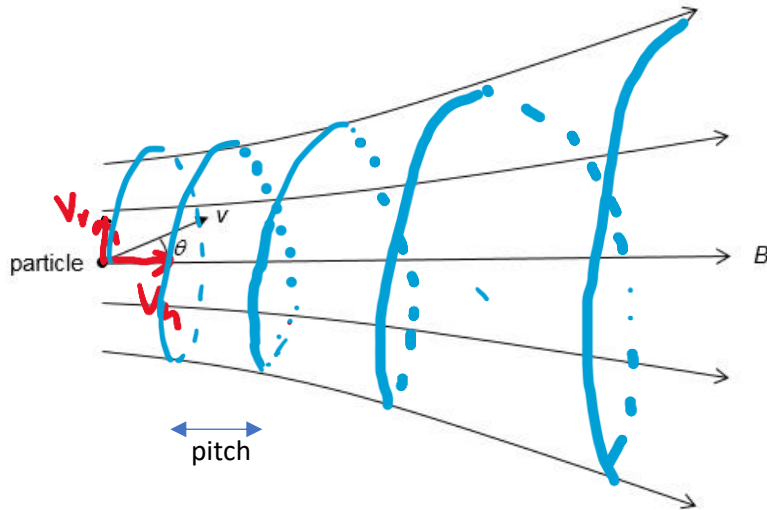


- 7 (a) (i) force =  $qE$  M1  
in the direction of the field A1
- (ii) no force B1  
(since current/velocity is parallel to B-field)
- (b) (i) B field pointing into the page/plane of the paper A1
- (ii) resultant force is zero/the 2 forces cancel out B1  
(good to show  $F_E$  acts downward and  $F_B$  acts upwards)  
 $Eq = Bqv \Rightarrow v = E/B$  B1

(c) (i)



- Horizontal component of velocity experience zero force, so constant horizontal motion to the right. B1
  - Vertical component of velocity results in a magnetic force into page which provides centripetal force for circular motion into the page B1
  - Since radius  $r = mv/Bq$ , as  $B$  decreases, radius  $r$  increases B1
- (ii) Helical path drawn with increasing radius and longer pitch B1  
(Note: The period of each circle  $T = 2\pi m/Bq$ , as  $B$  decreases  $T$  increases, so horizontal pitch  $s_x = v_x T$  increases)

